Bio Informatics

Lecture 11

Scoring System using Matrices

- Sequence alignment and database searching programs compare sequences to each other as a series of characters.
- All algorithms (programs) for comparison rely on some scoring scheme for that.
- Different scoring systems are used to assign a score to each comparison of a pair of characters.
- In most cases a positive score is given to identical or similar character pairs, and a negative or zero score to dissimilar character pairs.

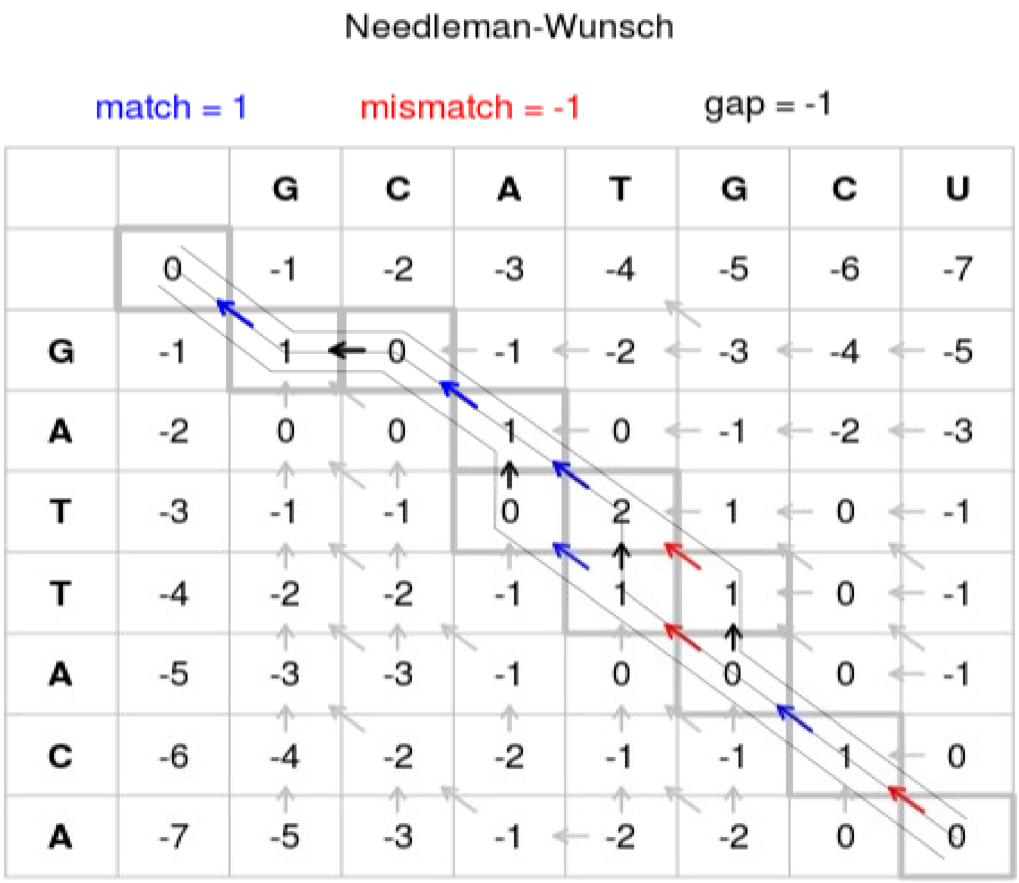
Scoring Systems using Matrices

Needleman-Wunsch

match = 1 mismatch = -1 gap = -1

		G	C	A	T	G	C	U
	0	-1	-2	-3	-4	-5	-6	-7
G	-1	1	0	-1	-2	-3	-4	-5
A	-2	0	0	1	0	-1	-2	-3
T	-3	-1	-1	0	2	1	0	-1
T	-4	-2	-2	-1	1	1	0	-1
A	-5	-3	-3	-1	0	0	0	-1
C	-6	-4	-2	-2	-1	-1	1	0
A	-7	-5	-3	-1	-2	-2	0	0

Scoring using Matrices



	A	C	G	T
A	1	-1	-1	-1
C	-1	1	-1	-1
G	-1	-1	1	-1
T	-1	-1	-1	1

Scoring using Matrices

Match: +1; Mismatch: -1; Gap: -1;

		C	O	E	L	A	C	A	N	T	H
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
E	0	0	0	1	0	0	0	0	0	0	0
L	0	0	0	0	2	1	0	0	0	0	0
I	0	0	0	0	1	1	0	0	0	0	0
C	0	1	0	0	0	0	2	1	0	0	0
A	0	0	0	0	0	1	0	3	2	1	0
N	0	0	0	0	0	0	0	2	4	3	2

Scoring System using Matrices

- In most cases of assigning scores to the specific alignments, we have seen that the scores cannot be uniformed for all the matches, mismatches and gap penalties.
- As we know that due to variations in genetic mutations, several amino acids are easily replaced/changed by others without affecting the overall function of the protein.
- While others are not that easily replaced and cost very much in the functionality of that protein.
- Moreover, several conditions exist where gaps can be introduced with ease but in some locations of DNA or protein gaps cannot be inserted or deleted easily.

Scoring System using Matrices

- So the fixed scoring systems for all replacements/substitutions in given DNA/protein sequences is not a good solution.
- As we have seen earlier that, certain nucleotides and amino acids have a higher propensity to match.
- Whereas some nucleotides and amino acids have a higher propensity to mismatch.
- So we should consider such facts before assigning uniform or same match and mismatch scores to any replacements or substitutions.

Scoring Systems using Matrices – BLOSUM62

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4